

Timestamp & Validation-Based Protocols -- DBMS

Definition: Timestamp Ordering Protocols ensure conflict serializability without using locks by assigning a unique monotonic timestamp $TS(T_i)$ to each transaction upon start and enforcing operation ordering according to timestamp sequence.

1. Detailed Technical Explanation

1. System Timestamps

Each data item X maintains two timestamps:

2. Basic Timestamp Ordering Rules

Read Operation Read(X) by Transaction T_i :

Write Operation Write(X) by Transaction T_i :

2. Validation-Based (Optimistic) Protocol

Used when read operations dominate write operations (low conflict rate). Execution is split into 3 phases:

+-----+	+-----+	+-----+
1. READ PHASE	---> 2. VALIDATION	---> 3. WRITE PHASE
Read & Write to	Check for conflict	Copy workspace to
private workspace	with committed Tx	database storage
+-----+	+-----+	+-----+

Validation Test Condition:

For all T_k committed before T_i :

3. Core Concepts & Memory Keywords

- * Monotonic Timestamps: Unique growing values (System clock or global counter).

4. Must-Write Points for Exams

- * Timestamp ordering protocols guarantee freedom from deadlocks because transactions are never made to wait indefinitely.

5. Quick Recall Flow

Compare $TS(T_i)$ with W- $TS(X)$ /R- $TS(X)$ -> Valid Proceed : Outdated Abort & Restart with New Timestamp